

SAÉ 3-03: Description et prévision de données temporelles – Cheat Sheet v3

This document provides a mathematically rigorous synthesis of time series analysis and forecasting as defined in the course 'SAÉ 3-03: Description et prévision de données temporelles'. It highlights essential formulas for decomposition, seasonality adjustment (CVS), autocorrelation, and forecasting.

1. Fundamental Components and Models

A time series (série chronologique) is a sequence of observations X_t where the temporal dependence between variables is the primary source of information. Classically, a series is decomposed into three components:

- 1. **Trend (Composante tendancielle/Trend):** m_t , the long-term persistent direction.
- 2. **Seasonality (Mouvement périodique/saisonnier):** s_t , phenomena repeating at regular intervals p .
- 3. **Residuals (Composante résiduelle/aléa):** e_t , irregular "noise" with a zero mean.

Composition Models

Model Type	Equation	Usage
Additive	$X_t = m_t + s_t + e_t$	Used when seasonal variation amplitude is constant regardless of the trend level.
Multiplicative	$X_t = m_t \times s_t \times e_t$	Used when seasonal variation amplitude is proportional to the trend level.
Mixed	$X_t = m_t \times s_t + e_t$	(Not studied in this course).

2. Model Selection Methods (Méthodes de choix du modèle)

Method	Procedure	Criteria for Additive Model	Criteria for Multiplicative Model
Band Method (Méthode de la bande)	Plot lines through maxima and minima.	Lines are approximately parallel.	Lines diverge or converge (not parallel).

Method	Procedure	Criteria for Additive Model	Criteria for Multiplicative Model
Profile Method (Méthode du profil)	Superimpose seasonal cycles.	Curves are approximately parallel.	Cycles accentuate (peaks/troughs widen).
Buys-Ballot Method	Linear regression of standard deviation s_r against the mean $\bar{y}$.	$s_r = a\bar{y} + b$ where $a \approx 0$.	$a \neq 0$ (Standard deviation is a function of the mean).

3. Trend Estimation (Analyse de la tendance)

Non-Parametric: Centred Moving Averages (Moyennes mobiles centrées)

For a series of size N and period p :

- If p is odd ($p = 2k + 1$):

$$m_{p,t} = \frac{1}{p} \sum_{i=-k}^k X_{t+i} = \frac{X_{t-k} + \dots + X_t + \dots + X_{t+k}}{p}$$

- If p is even ($p = 2k$):

$$m_{p,t} = \frac{1}{p} \left(\frac{X_{t-k}}{2} + \sum_{i=-k+1}^{k-1} X_{t+i} + \frac{X_{t+k}}{2} \right) = \frac{\frac{X_{t-k}}{2} + \dots + X_t + \dots + \frac{X_{t+k}}{2}}{p}$$

Parametric: Trend Line (Droite de tendance)

If the moving average reveals a linear trend, the regression line $y = \beta_1 x + \beta_0$ is calculated:

$$\beta_1 = \frac{\text{cov}(m_{p,t}, t)}{\text{var}(t)}; \quad \beta_0 = \bar{m}_{p,t} - \beta_1 \bar{t}$$

4. Seasonality and Seasonally Adjusted Series (CVS)

This section details the calculation of the Seasonally Adjusted Series (CVS - Série corrigée des variations saisonnières) and Residuals.

A. Additive Model (Modèle additif)

- **Trend estimate:** $\hat{m}_t$ (via moving averages or regression).

- **Raw seasonal coefficients:** $\hat{s}_j$ calculated as the arithmetic mean of $(X_t - \hat{m}_t)$ for each period j .
- **Normalised seasonal coefficients:** $s_j^* = \hat{s}_j - \bar{s}$, where $\bar{s} = \frac{1}{p} \sum_{i=1}^p \hat{s}_i$.
- *Property:* $\sum s_j^* = 0$.
- **Seasonally Adjusted Series (CVS):**

$$X_t^* = X_t - s_t^*$$

- **Additive Residuals:**

$$e_t = X_t^* - \hat{m}_t = X_t - \hat{m}_t - s_t^*$$

B. Multiplicative Model (Modèle multiplicatif)

- **Trend estimate:** $\hat{m}_t$.
- **Raw seasonal coefficients:** $\hat{s}_j$ calculated as the arithmetic mean of $\left(\frac{X_t}{\hat{m}_t}\right)$ for each period j .
- **Normalised seasonal coefficients:** $s_j^* = \frac{\hat{s}_j}{\bar{s}}$, where $\bar{s} = \frac{1}{p} \sum_{i=1}^p \hat{s}_i$.
- *Property:* Mean of $s_j^* = 1$.
- **Seasonally Adjusted Series (CVS):**

$$X_t^* = \frac{X_t}{s_t^*}$$

- **Multiplicative Residuals:**

$$e_t = \frac{X_t^*}{\hat{m}_t} = \frac{X_t}{\hat{m}_t \times s_t^*}$$

5. Autocovariance and ACF (Indices de dépendance)

These metrics identify temporal dependence and stationarity (stationnarité).

- **Autocovariance function:**

$$\gamma(k) = \mathbb{E}[(X_t - \mu)(X_{t+k} - \mu)]$$

$$\text{Estimator: } \hat{\gamma}(k) = \frac{1}{N} \sum_{t=1}^{N-k} (X_t - \bar{X})(X_{t+k} - \bar{X})$$

- **Autocorrelation Function (ACF):**

$$\rho(k) = \frac{\gamma(k)}{\gamma(0)}$$

$$\text{Estimator: } \hat{\rho}(k) = \frac{\hat{\gamma}(k)}{\hat{\gamma}(0)}$$

- **Properties:**

- $\rho(0) = 1$.

- Symmetry: $\rho(k) = \rho(-k)$.
- Bounds: $-1 \leq \rho(k) \leq 1$.
- **White Noise (Bruit blanc):** $\rho(k) \approx 0$ for all $k \neq 0$.

6. Statistical Tests (Tests sur les séries)

Test	Null Hypothesis (H_0)	Alternative Hypothesis (H_1)	Decision Rule
Augmented Dickey-Fuller (ADF)	Series is non-stationary.	Series is stationary.	Reject H_0 if $p\text{-value} < 0.05$.
Mann-Kendall	No trend exists.	Monotonic trend exists.	If $p\text{-value} < 0.05$ and $\tau > 0$, trend is increasing.
Pettitt	No change in central trend.	Presence of a rupture point.	Identifies date of shift in mean.
KPSS	Series is stationary.	Series is non-stationary.	Reject H_0 if $p\text{-value} < 0.05$.

7. Forecasting Methods (Prévision)

Parametric Modelling

A forecast $\hat{x}_{T+h}$ for horizon h is built by combining trend and seasonality:

- **Additive:** $\hat{x}_{T+h} = \hat{m}_{T+h} + \hat{s}_{T+h}$
- **Multiplicative:** $\hat{x}_{T+h} = \hat{m}_{T+h} \times \hat{s}_{T+h}$, where $\hat{m}_{T+h} = a(T+h) + b$.

Exponential Smoothing (Lissage exponentiel)

1. **Simple Exponential Smoothing (LES/SES):** For series with no trend or seasonality. Update:
 $\hat{x}_T(h) = \beta x_T + (1 - \beta)\hat{x}_{T-1}(1)$, where β is the smoothing constant.
2. **Holt's Double Exponential Smoothing (LED):** For series with a local linear trend but no seasonality.

$$\hat{a}_T = \hat{a}_{T-1} + (1 - \beta)^2(x_T - \hat{x}_{T-1}(1))$$

3. **Holt-Winters (HW):** For series with trend and seasonality. **Additive Update Formulas:** - Level:
 $\hat{a}_T = \beta(x_T - \hat{s}_{T-p}) + (1 - \beta)(\hat{a}_{T-1} + \hat{b}_{T-1})$ - Trend: $\hat{b}_T = \alpha(\hat{a}_T - \hat{a}_{T-1}) + (1 - \alpha)\hat{b}_{T-1}$ - Seasonality: $\hat{s}_T = \gamma(x_T - \hat{a}_T) + (1 - \gamma)\hat{s}_{T-p}$

8. Validation Metrics (Indicateurs d'erreur)

To evaluate model quality, calculate the global magnitude of residuals $e_t = x_t - \hat{x}_t(1)$:

Metric	Formula	Terminology
Mean Error	$EM = \frac{1}{N} \sum_{i=1}^N (x_{i+1} - \hat{x}_i(1))$	Erreur Moyenne
Mean Absolute Error	$EAM = \frac{1}{N} \sum_{i=1}^N x_{i+1} - \hat{x}_i(1) $	Erreur Absolue Moyenne
Mean Squared Error	$EQM = \frac{1}{N} \sum_{i=1}^N (x_{i+1} - \hat{x}_i(1))^2$	Erreur Quadratique Moyenne

Note: For a well-adjusted model, EM should be close to 0 and the EQM should be minimised.